The National Hospital for Neurology and Neurosurgery
Neuroscience Critical Care Unit (NCCU)

University College London Hospitals NHS
NHS Foundation Trust

Management of Sodium and Water Balance
HYPERNATRAEMIA (Na+ >145mmol/L)

CONSIDER TREATING SERUM SODIUM ≥155mmol/L

Normal plasma sodium =135–145mmo/L & plasma osmolality 285–295mOsm/kg

Sodium disturbances are common after brain injury, are often self-limiting and patient asymptomatic. Treatment is indicated when there are acute symptoms that pose a risk of significant neurological complications and possibly death.

The symptoms and signs relate not only to the severity of the sodium disturbance but also to the rate of change of serum sodium concentration.

HYPERNATRAEMIA is defined as a serum sodium concentration of >145mmol/L. In the context of brain injury, diagnosis of DI is made if there is:

a) increased urine volume (>3000mL/24h)
b) high serum sodium (>145mmol/L)
c) high serum osmolality (>305mmol/kg)
d) abnormally low urine osmolality (<350mmol/kg)

HYPERNATRAEMIA is classified as:

MILD	146-149mmol/L
MODERATE	150-169mmol/L
SEVERE	≥170mmol/L

SIGNS and SYMPTOMS include:

MODERATE	SEVERE
Lethargy Thirst Irritability Restlessness Muscle weakness Confusion	Ataxia Hypereflexia Seizures Coma Death

Bradshaw, K & Smith, M: Continuing Education in Anaesthesia, Critical Care & Pain j Volume 8 Number 4 2008

DIABETES INSIPIDUS (DI)

Can occur in the following:

- Severe TBI or major cerebral insult (*due to generalised brain swelling)*
- Pituitary dysfunction - transphenoidal surgery / craniopharyngioma
- Brain stem death

SIGNS

A high urine output, dilute in appearance and with SG 1.005, and a high serum sodium is suggestive of DI and should prompt further laboratory testing - paired urine and serum osomolarities

NORMAL urinary sodium	20-60mmol/L
HIGH serum osmolality	>305mmol/kg
LOW urine osmolality	<350mmol/kg

- Low CVP (<5mmHg)
- High urine output **(>1000mL in 4h)** – *In patients awaiting brain stem testing **do not wait** until urine output reaches this prescribed volume before giving DDAVP – treat early when Na+ is higher than normal range*
- Low specific gravity **(≤1.005)** - *Consider DDAVP even if SG normal*

TREATMENT - *Aim to conserve fluid volume and electrolytes by reducing urine output*

- Serum sodium concentration should **not** be reduced by **>10-12mmol/L/day**
- GIVE 0.2 or 0.4 mcg DDAVP IV (*Vial 4mcg/mL – dilute to 10mL = 0.4mcg/mL)*
- Repeat after 30min if urine output remains high
- **Only** use 5% dextrose if:
  - Na^+ ≥155mmol/L
  - Consultant-led decision
  - Risks of brain swelling in severe brain injury in acute phase considered
- Keep blood glucose 7-12mmol/L
- Consider NG water for patients with NGT

POST PITUITARY SURGERY

- Patients usually require IV DDAVP in 1st 48h post-op
- Patients often very thirsty with high urine output
- Allow to drink freely & give IV DDAVP as above
- Do not chase urine output
- Input from Endocrinology Team
- Intranasal / oral DDAVP after acute episode for patients who require chronic replacement

DEHYDRATION

Can occur in the following:

- Renal water loss *(diuretic drugs)*
- Extrarenal water loss *(vomiting / diarrhoea / fever)*
- Reduced water intake / poor thirst

SIGNS

- Low CVP (<5mmHg)
- Low urine output (<0.5mL/kg/h)
- High specific gravity (≥1.020)
- Tachycardia and hypotension
- 'Dry' on clinical assessment

TREATMENT - *Aim to establish normal urine output by normalising fluid volume and electrolytes*

- Serum sodium concentration should ***not*** be reduced by **>10-12mmol/L/day**
- Use **isotonic fluids** to restore normovolaemia
- Increase fluid intake if indicated with careful assessment of individual patient requirements
- Consult with dietician if patient at risk of refeeding syndrome

MONITOR

- ☐ CARDIOVASCULAR STATUS *(risk of instability due to intravascular volume depletion and loss of electrolytes)*
- ☐ GCS
- ☐ FLUID BALANCE
- ☐ SERUM SODIUM
- ☐ RENAL FUNCTION
- ☐ URINE AND SERUM OSMOLALITY *(at least BD until stable)*